Risk-Weighted apoB Outperforms Unweighted apoB or Lipoprotein(a) as a Biomarker of Residual Cardiovascular Risk After Acute Coronary Syndrome and the Benefit of Alirocumab Therapy

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Abstract:
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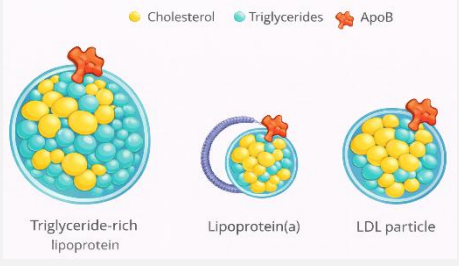
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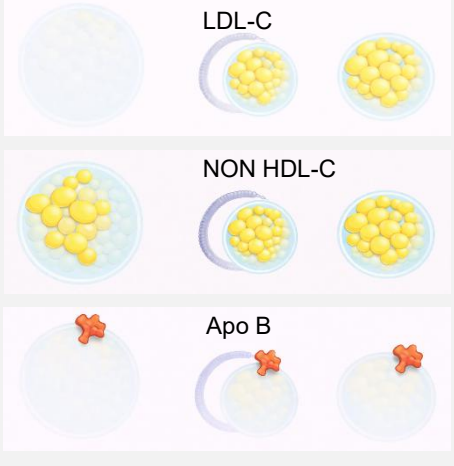
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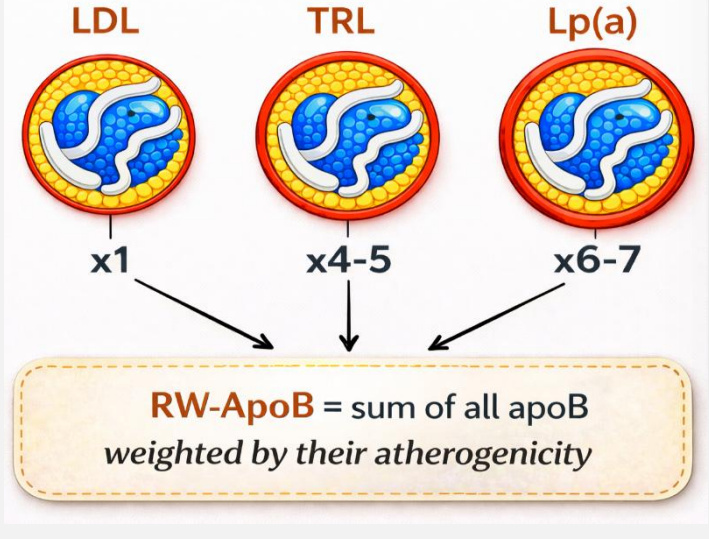
RISK-WEIGHTED APOB: CONCEPT

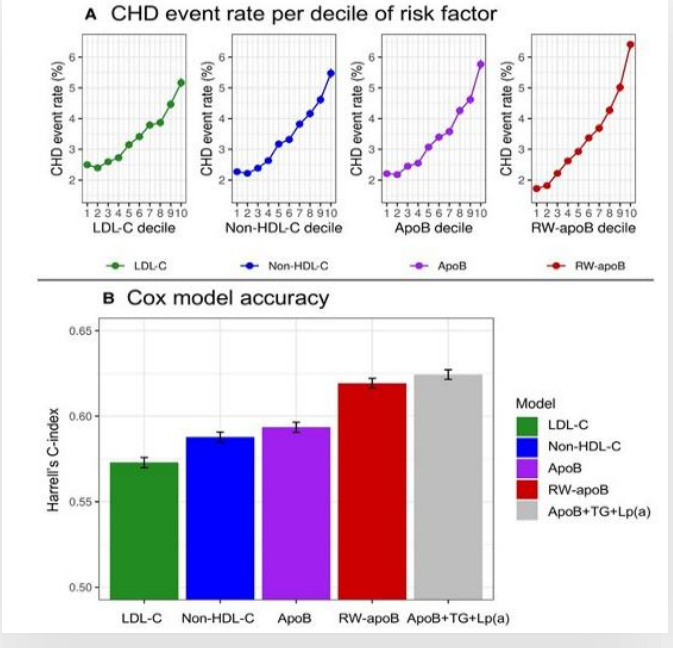
A simple, integrated metric of the atherogenic risk of apoB-containing lipoproteins





ApoB particles
are not
equally atherogenic





Primary prevention, not on lipid lowering therapy

Figure 3, Rehman MB & Björnson E et al. EHJ 2026

Objectives and Methods

Compared to ApoB or Lp(a), does RW-apoB:

1. Improve prediction of residual risk on statins?

2. Better identify treatment benefit from PCSK9 inhibition?

RW-apoB (mg/dL)
=
0.132 x TG (mg/dL) **+**
0.215 x Lp(a) (nmol/L) **+**
0.736 × apoB (mg/dL)

Post hoc analysis ODYSSEY OUTCOMES trial

18,924 **recent ACS and elevated atherogenic lipoproteins despite optimized statin therapy** → randomized alirocumab vs placebo → 11,957 with Lp(a) in nmol/L→ median follow-up 2.8 y

PRIMARY OUTCOME (MACE):
coronary heart disease death + nonfatal myocardial infarction + fatal/non-fatal ischemic stroke + hospitalization for unstable angina

STATISTICS:

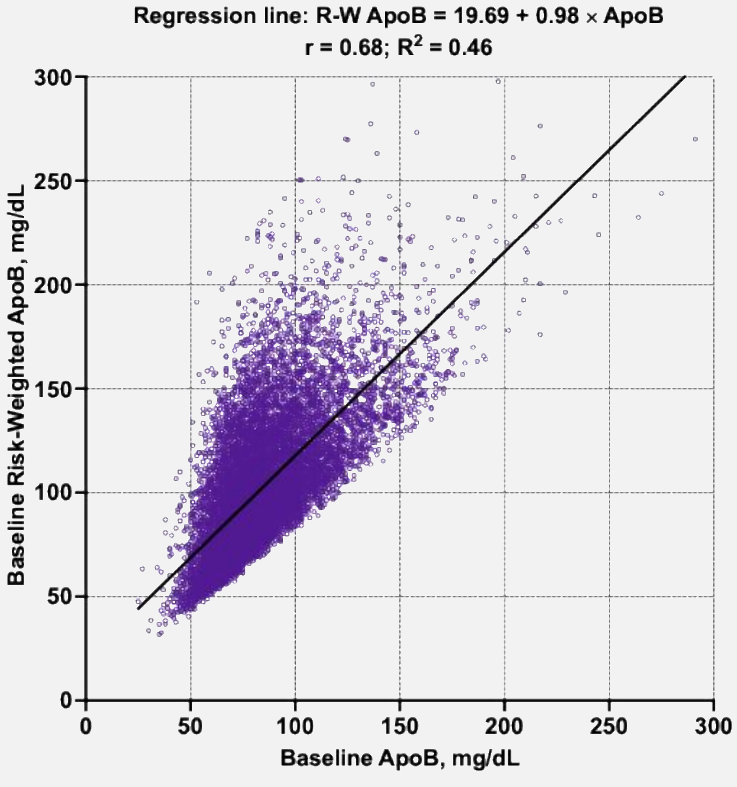
Placebo arm: Cox models

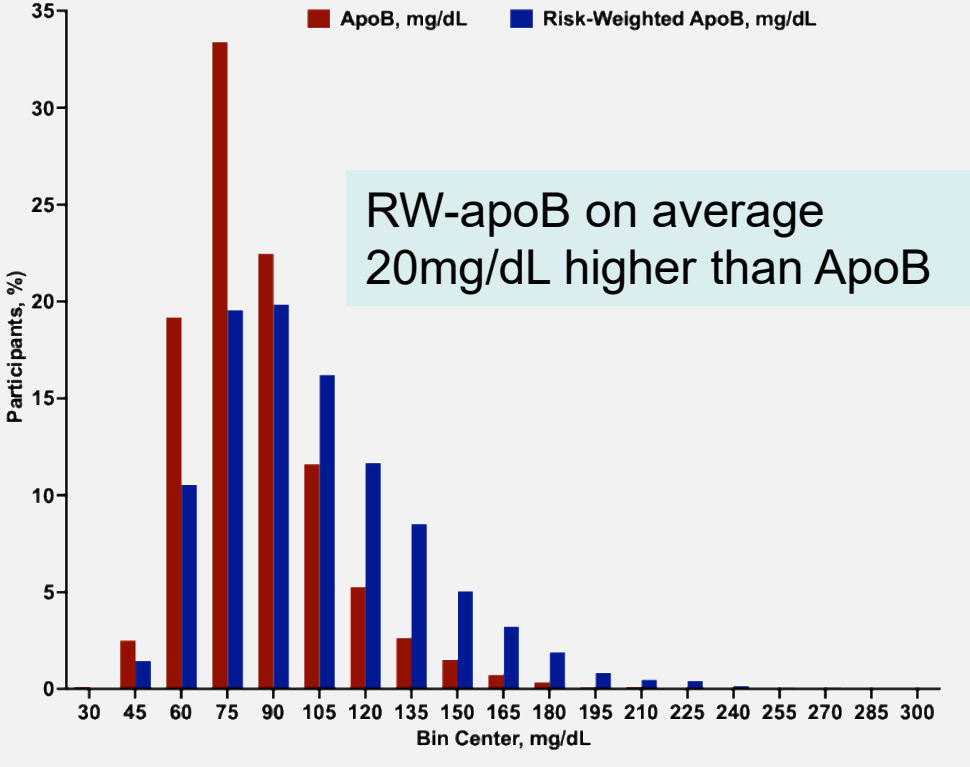
- Univariable: do baseline Lp(a), apoB and RW-apoB predict **residual CV risk**?
- Multivariable: do apoB or Lp(a) provide additional prediction beyond RW-apoB?

Both arms: Spline from unadjusted Cox models

- relationship between baseline biomarkers (percentiles) and CV risk and **absolute risk reduction** on alirocumab.

RESULTS: RW-APOB VS APOB





RW-apoB on average 20mg/dL higher than ApoB

Median baseline values: Lp(a) 45,0nmol/L LDL-C 86,5mg/dL ApoB 80,0mg/dL TG 131mg/dL

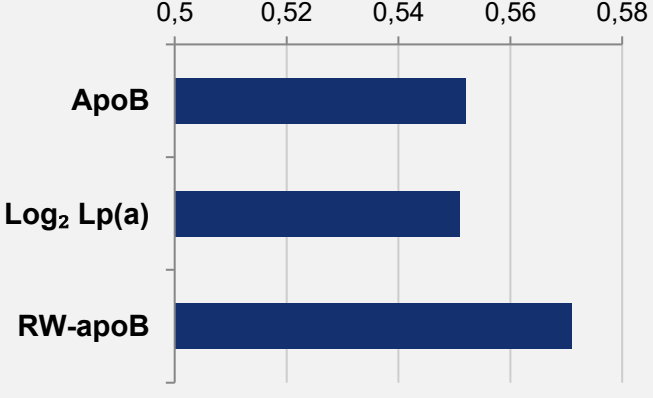
RESULTS: RW-APOB BEST IDENTIFIES RESIDUAL RISK IN THE PLACEBO ARM

UNIVARIABLE MODELS	Predictor	P value
	ApoB	<0.0001
	Log_2 Lp(a)	<0.0001
	RW-apoB	<0.0001
MULTIVARIABLE MODELS	**Predictor**	**P value**
RW-apoB + ApoB	**RW-apoB**	0.0001
	ApoB	0.0455
RW-apoB + Log_2 Lp(a)	**RW-apoB**	<0.0001
	Log_2 Lp(a)	0.3091

In multivariable Cox models, RW-apoB remained associated with events after adjusting for apoB (P≤0.0001)

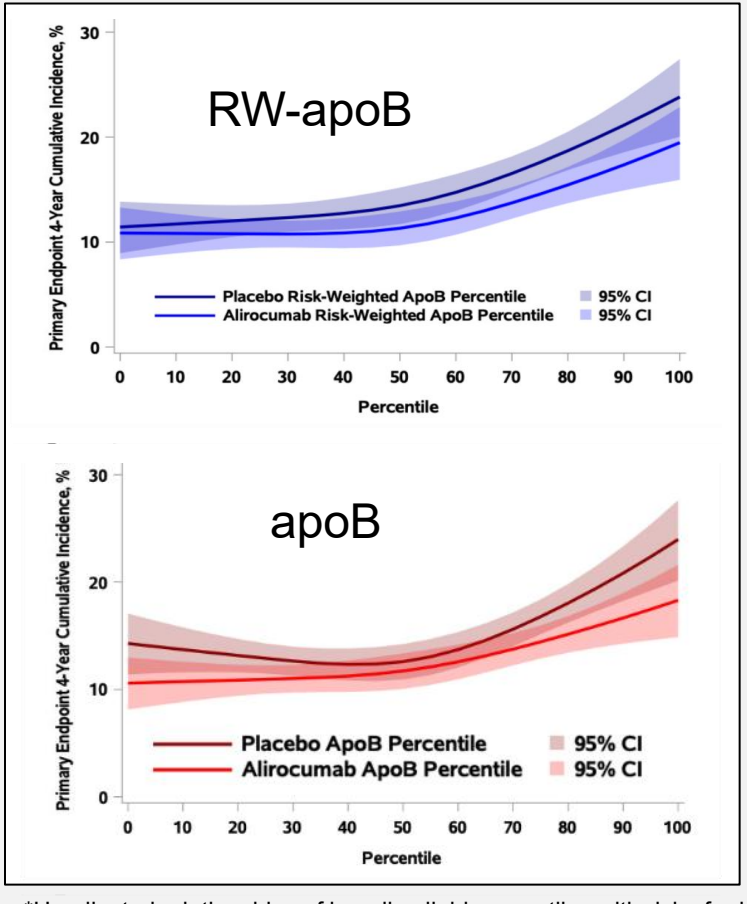
Lp(a) provided no additional prognostic information beyond RW-ApoB

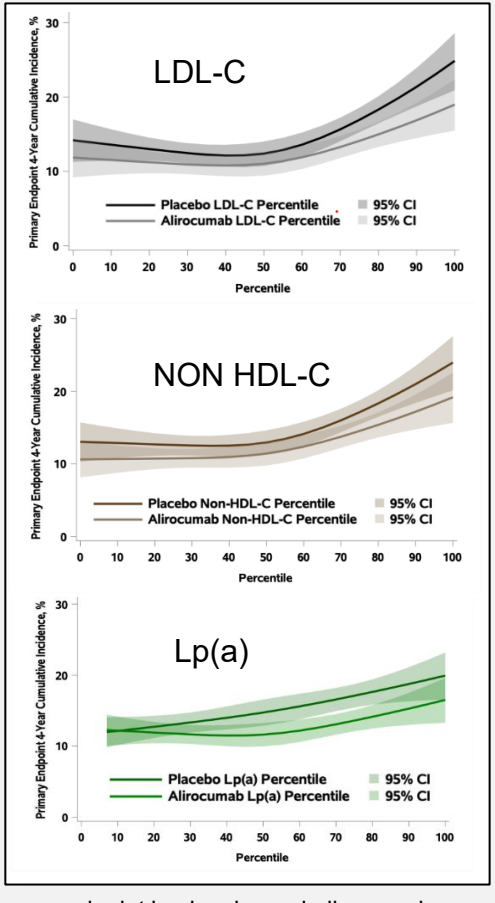
RW-apoB identified residual CV risk better than apoB or Lp(a)



DISCRIMINATION (C-INDEX)

RESULTS: RISK-WEIGHTED APOB BEST IDENTIFIES TREATMENT BENEFIT*





- **In the placebo group**, RW-apoB gave a steep and monotonic gradient of MACE risk across its full range, compared with apoB and other lipids.
- In the alirocumab group, RW-apoB demonstrated a continuous gradient of **treatment benefit** compared with apoB, LDL-C, and non-HDL-C.
- These favorable characteristics of the RW-apoB relationships largely reflect the **weighting of Lp(a)** in RW-apoB.

*Unadjusted relationships of baseline lipid percentiles with risk of primary endpoint in placebo and alirocumab groups

Conclusion

In patients with recent ACS and elevated lipoproteins despite optimized statin therapy, RW-apoB outperforms apoB and Lp(a)

- in predicting residual risk in the placebo arm

- in identifying the treatment benefit of alirocumab

RW-apoB online calculator at **www.rwapob.com**